

Part A: Binary Segmentation with U-Net

Oxford-IIIT Pet Dataset - candidate technical assessment. Prepared 30 September 2026.

Item	Value
Task	Binary pet-vs-background segmentation
Dataset	Oxford-IIIT Pet (images + trimap annotations), Kaggle mirror by julinmalooof
Usable pairs	7,367 of 7,390 (23 excluded: confirmed broken/empty ground-truth labels)
Split	70/15/15, fixed seed 42: 5,156 train / 1,105 val / 1,106 test
Model	U-Net from scratch, 4 encoder levels + bottleneck, 7,765,985 parameters
Best result (Exp C)	Test Dice 0.9028, Test IoU 0.8320 (BCE+Dice loss + augmentation)
Experiments run	3 controlled, one-factor-at-a-time (loss function, then augmentation)

Headline findings

- Each controlled change improved test Dice: BCE alone 0.8937 -> BCE+Dice 0.8969 (+0.0032) -> BCE+Dice+augmentation 0.9028 (+0.0091 total from baseline).
- Val and test scores track closely across all three experiments (e.g. Exp A: val 0.8902 vs test 0.8937), indicating the split and evaluation are behaving correctly, not overfit to validation.
- Dice and IoU implementations were unit-tested against hand-computed cases (Dice=1/IoU=1 for a perfect prediction; Dice=0.667/IoU=0.5 for a known half-overlap case) before any training, to rule out a metric bug being mistaken for model learning.
- The dataset audit caught two real data-quality issues before modelling: 23 trimaps with zero foreground pixels despite a clearly visible pet in the photo (excluded), and 12 images in non-RGB color modes (handled by forcing RGB at load time).
- Per-image error analysis on the best model surfaced 3 distinct, explainable failure patterns (below), including a second, independent case of a corrupted ground-truth label caught by inspecting the single worst test-set prediction.

1. Dataset and audit

Oxford-IIIT Pet: 7,390 image/trimap pairs across 37 breeds (~200 images per breed, std 1.48 - well balanced). Trimap pixel values are the standard {1=pet, 2=background, 3=border/ambiguous}, confirmed across all 7,390 masks before any processing.

Check	Result
Image-mask pairing	7,390 images and 7,390 trimaps, 0 orphaned on either side, 0 duplicates
Corruption	0 unreadable images, 0 unreadable trimaps
Trimap values	Exactly {1,2,3} across all 7,390 masks - no unexpected values
Image color modes	RGB 7,378 P (palette) 6 RGBA 3 L (grayscale) 3 -> forced to RGB at load time
Image size	Width 114-3,264px (mean 437), height 103-2,606px (mean 391) - natural photos, resized to 256x256 for training
Pixel-class balance (sampled)	Foreground 29.3%, background 58.7%, border 11.9% - motivates Dice/IoU over pixel accuracy
Empty-foreground masks	23 of 7,390 (0.31%) - trimap has zero pet pixels despite a visible pet in the photo

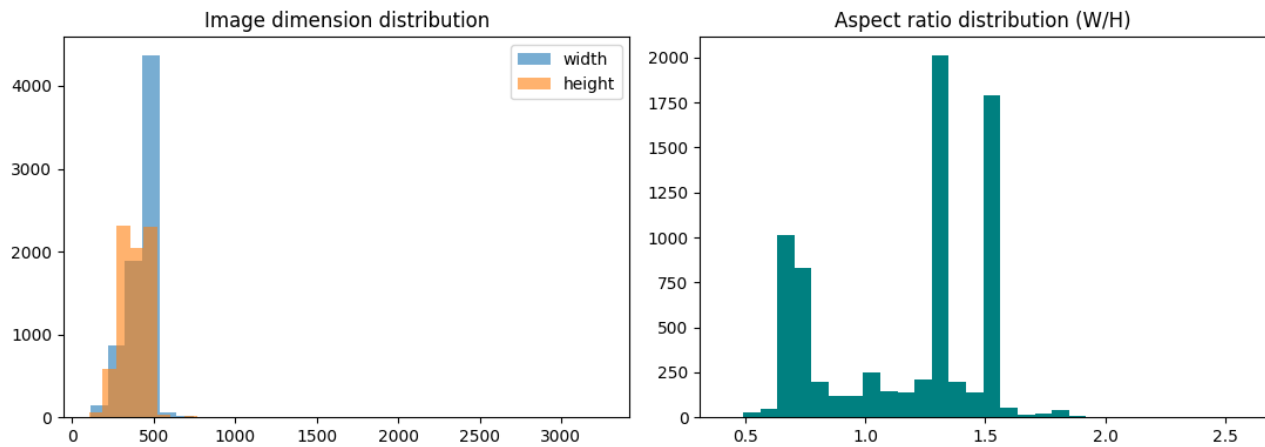
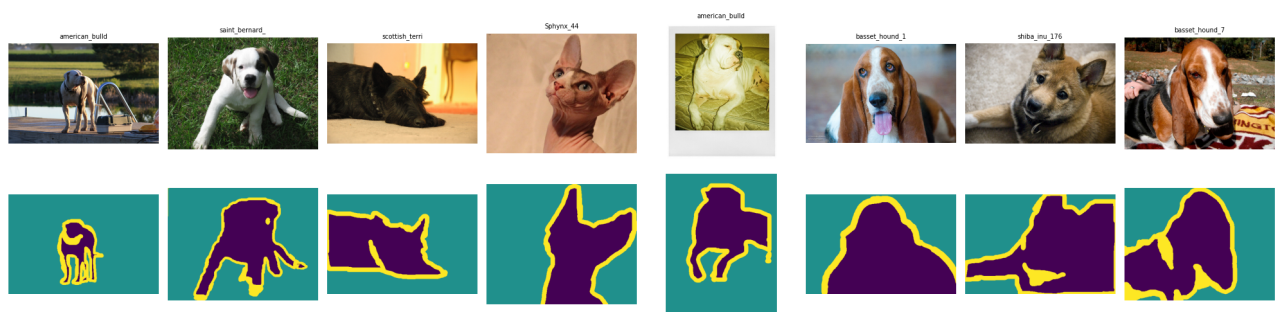


Image and mask dimension / aspect-ratio distributions across the full dataset.



8 image/trimap pairs sampled from different parts of the dataset, confirming correct pairing before any modelling.

Excluded: 23 broken-label images

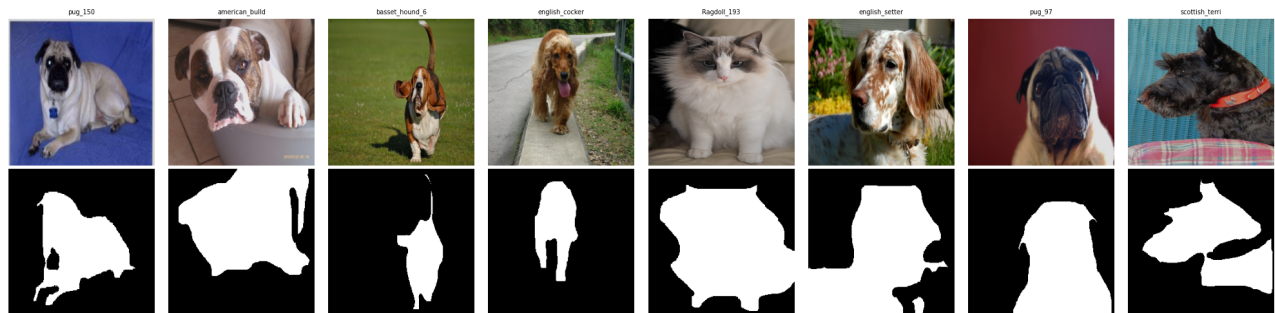
All 23 stems with zero foreground pixels were inspected visually. In every case a pet is clearly visible and centred in the photo, yet the trimap is solid background. 10 of the 23 are Egyptian_Mau, suggesting a batch labelling error for that breed rather than 23 independent edge cases. These 23 were excluded from train/val/test entirely (7,390 -> 7,367 usable pairs), and the exclusion is applied before splitting, seeded, and reproducible.



All 23 empty-mask cases: photo (top) vs trimap (bottom). Every trimap is solid background despite an unmistakable pet in the photo.

2. Preprocessing and split

Step	Rule and reason
Color mode	Every image forced to RGB at load time (fixes the 12 non-RGB images found in the audit)
Image resize	256x256, bilinear (safe for continuous pixel values)
Mask resize	256x256, nearest-neighbour only - bilinear would blend class labels 1/2/3 into invalid fractional values
Trimap -> binary mapping	foreground(1) -> 1; background(2) AND border(3) -> 0 (documented, conservative default for the binary task)
Normalization	Images scaled to [0,1] by /255; no further channel normalization applied
Augmentation (train split only)	Horizontal flip (p=0.5), brightness jitter (0.8-1.2x, p=0.5), contrast jitter (0.8-1.2x, p=0.5). Flip applied identically to image and mask; brightness/contrast to image only
Split	70/15/15, sklearn stratified split, seed 42, applied to the 7,367 usable pairs -> 5,156 / 1,105 / 1,106



A post-preprocessing training batch. Top: augmented images (note the flip). Bottom: matching binary masks - solid black/white with no gray blur, confirming nearest-neighbour resizing preserved valid label values. Verified programmatically: mask tensor unique values were exactly {0.0, 1.0}.

3. Model: U-Net baseline

Built from scratch (not a pretrained backbone) as the baseline architecture. 4 encoder levels (32/64/128/256 channels) + a 512-channel bottleneck, with a symmetric decoder that concatenates the matching encoder feature map at each level (skip connections) before refining with a double-convolution block. Output: a single-channel logit map at the input resolution, passed through a sigmoid at inference/evaluation time.

Stage	Output shape	Note
Input	(B, 3, 256, 256)	RGB image, batch B
enc1 (encoder, level 1)	(B, 32, 256, 256)	No downsampling yet at this level
enc2 (encoder, level 2)	(B, 64, 128, 128)	Resolution halved, channels doubled
Bottleneck	(B, 512, 16, 16)	Most compressed / most abstract representation (256 / 2^4)
up4 (decoder, first upsample)	(B, 256, 32, 32)	Before concatenation with the matching encoder skip
Output	(B, 1, 256, 256)	Matches input resolution exactly, single channel

Total parameters: **7,765,985** (all trainable). All four concatenation points matched shapes exactly with no cropping required, confirmed by a forward pass with random input before any real training.

4. Loss functions and metrics - unit-tested before training

Dice = $(2|P \text{ intersect } G| + \text{eps}) / (|P| + |G| + \text{eps})$; IoU = $(|P \text{ intersect } G| + \text{eps}) / (|P \text{ union } G| + \text{eps})$. Both computed per image, then averaged over the batch (not a single flattened score), per the requirement to report mean-per-image Dice/IoU. eps prevents a divide-by-zero on an empty prediction or an empty ground truth.

Test case	Expected	Got	Pass
Perfect prediction	Dice=1.0000, IoU=1.0000	Dice=1.0000, IoU=1.0000	Yes
Empty prediction (no eps crash)	Dice=0.0000, IoU=0.0000	Dice=0.0000, IoU=0.0000	Yes
Completely wrong (opposite) prediction	Dice=0.0000, IoU=0.0000	Dice=0.0000, IoU=0.0000	Yes
Known half-overlap (16 true px, 8 pred, 8 overlap)	Dice=0.6667, IoU=0.5000	Dice=0.6667, IoU=0.5000	Yes
Dice >= IoU property (mathematical identity)	True on every case	True	Yes
BCE+Dice loss on perfect vs opposite	near 0 vs high	0.0001 vs 5.5000	Yes

BCE+Dice loss combines nn.BCEWithLogitsLoss with (1 - soft Dice), weighted 0.5/0.5. BCE gives a stable, smooth per-pixel gradient early in training; Dice loss directly optimizes the overlap metric the model is evaluated on. All loss and metric code was verified against known-correct cases (above) before being used in any real training run.

5. Controlled experiments

Three experiments, each changing exactly one factor from the previous one, per the required one-factor-at-a-time design. Same U-Net architecture, same seed, same split, same optimizer (Adam) and initial learning rate (1e-3) with ReduceLROnPlateau, same checkpoint rule (best validation Dice, not lowest loss, since Dice is the task metric) in every experiment.

Experiment	Change from previous	Val Dice (best)	Test Dice	Test IoU	Train time
A - Baseline	BCE loss, no augmentation	0.8902	0.8937	0.8181	57.8 min
B - Loss change	BCE+Dice loss (aug still off)	0.8942	0.8969	0.8237	57.9 min
C - Augmentation added	BCE+Dice + flip/brightness/contrast aug	0.8969	0.9028	0.8320	57.9 min

- Val and test Dice track closely in every experiment (largest gap: Exp A, val 0.8902 vs test 0.8937 - test slightly higher, not lower, so no sign of overfitting to validation).
- Loss change alone (A -> B): +0.0032 test Dice, +0.0056 test IoU. Small but consistent - Dice loss directly targets the overlap metric, BCE does not.
- Augmentation on top of B (B -> C): a further +0.0059 test Dice. Experiment C also ran the full 30 epochs without early stopping (patience=7 never triggered), while A stopped improving earlier - Exp C's training loss stayed higher throughout (0.105 vs 0.069 final) while validation Dice was similar-or-better, the expected signature of augmentation working as intended rather than just adding noise.
- Combined effect, baseline to best: +0.0091 test Dice, +0.0139 test IoU. Neither individual change was dramatic; the gains stacked in the same direction across three independent runs.

6. Qualitative and error analysis (best model: Experiment C)

Per-image test Dice was computed and ranked for all 1,106 test images (not just the batch-average). Median (0.9267) exceeds the mean (0.9028), confirming a small number of outliers pull the average down rather than a systemic weakness: only 7/1,106 (0.6%) score below Dice 0.5, and 32/1,106 (2.9%) below 0.7.



12-image grid from the ranked test set: rows 1-4 worst, 5-8 median, 9-12 best. Columns: input image, ground-truth mask, predicted mask, overlay (green=correct foreground, red=missed pet pixel, blue=false-positive pixel).

#	Failure mode	Example	Likely cause	Confidence
1	Ground-truth label error	Maine_Coon_268 (Dice=0.000)	Full-resolution inspection shows a plausible, roughly correctly-placed prediction against what appears to be an inverted/incorrect trimap - a second, independently-found case of the same label-quality issue as the 23 excluded empty masks	High - visually clear on inspection
2	Human co-occurrence	Photo of a person holding a cat	The model's foreground blob lands on the human's head/hair rather than the small cat; trimaps never label humans, so any photo with a prominent human is a near-guaranteed hard case	High
3	Tiny/ambiguous subject in clutter	Dark mirror-selfie photo, cat small in frame	Pet occupies a very small, low-contrast fraction of a busy, dark scene	Medium
4	Low subject/background contrast	White Persian cat on a light background	Model over-predicts (false positives) when foreground and background tones are similar, likely relying partly on intensity contrast at the true boundary	Medium

7. Limitations

- Single training run per experiment (one seed each) - differences between A/B/C, while consistent in direction, have not been confirmed with repeated runs or a significance test.
- Binary threshold fixed at 0.5 for all reported Dice/IOU; not swept or tuned on the validation set.
- Augmentation strength (flip + +/-20% brightness/contrast) was a fixed, untuned choice - not compared against stronger or different augmentation policies.
- Border-class trimap pixels (11.9% of the average mask) were folded into background rather than modelled separately or excluded from the loss; this is a documented but untested design choice.
- No pretrained encoder was used for Part A (U-Net trained from scratch) - an optional stronger-model experiment was not run.
- The 23 excluded and 1 additional suspected-corrupt label (Maine_Coon_268) were found by manual/visual inspection, not a systematic label-quality scan of the full dataset - other undetected label errors may remain.

8. What I would try next

- A pretrained encoder (e.g. ResNet-34 or MobileNetV2 backbone) in place of the from-scratch encoder, to test whether transfer learning improves Dice further, especially on the low-contrast and cluttered-scene failure cases.
- Threshold tuning on the validation set (sweep 0.3-0.7) rather than a fixed 0.5, since Dice/IOU are threshold-sensitive metrics.
- A systematic full-dataset label-quality scan (e.g. flagging trimaps whose foreground area is a statistical outlier for their breed) to catch label errors beyond the 24 found so far.
- Targeted augmentation (stronger color/contrast jitter) aimed specifically at the low-contrast failure mode identified in the qualitative review.